PlanArena: an exact, learning-assisted land-use optimizer with empirical evaluation

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Statements and Declarations

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Data availability. All inputs are public and are downloaded by script; a manifest records each input's URL, retrieval date and SHA-256. The result file behind every number, the figures and each city's ingestion report are in the repository: `https://github.com/ParyaPayami/PlanArena`, release `v1.1-review`. [Archived DOI (Zenodo) to be added by the author.]

Code availability. The code is released under the MIT licence at `https://github.com/ParyaPayami/PlanArena`. An online demonstration of the Studio (Manhattan) is at `https://planarena-manhattan.onrender.com` and the game at `https://planarena-manhattan.onrender.com/play`.

Author contributions (CRediT). Parya Payami: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing (original draft), writing (review and editing).

Use of AI tools. Claude (Anthropic), used through Claude Code, assisted in writing and testing the code, running the analyses, and drafting the text and Online Resource 1. The author designed the study and the system, reviewed the code, results and text, read the cited sources and is accountable for the content. No AI system generated data or results. This is also stated in the Methods section of the manuscript.